Competent

97/100

97%

Module #	671101000-KM-01
NQF Level	level 4,
Notional hours	50
Credit(s)	Cr 5
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

FORMATIVE ASSESSMENT

HEALTH, SAFETY, QUALITY AND LEGISLATION

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions. • Ensure that all questions are completed. • Ensure that the completion of the workbook is their own work. • Ensure that all annexures are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

Special arrangements for assessment

Place	EastC Electrical
Language	English
Resources	
Barriers	N/A

People to be involved with assessment

Learner: Sela	Manager:
Trainer: Miss R Nyakala	Mentor / Coach:
Facilitator Miss R Nyakala	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Sela Phiri

Learner's signature: S Date

Facilitator Name: Miss R Nyakala

Facilitator's signature: Date

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

S

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: S. _____ Date: _____

Employee witness: _____ Date: _____

S

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Sela Phiri	Signature S
Facilitator Name: Miss R Nyakala	Signature:

Complete the following questions as per instructions provided

S

- Hand Protection (Insulated Rubber Gloves / leather Gauntlets) insulated gloves protect against electric shock
- eye Protection (safety Glasses / Face Shield) protect eys From flying debris, Chemical Splashes, or electrical arc Flashes
- Foot protection (Steel-toe Safety Boots) protects Feet From heavy Falling objects and provides electrical hazard (EH) rated Soles For insulation
- Body Prot Protection (Flame-Resistant) Overalls (PPE) protect protects the skin From heat and burns caused by arc Flashes or Fire
- hearing protection: Earplugs, earmuffs (loud noises)
- Respiratory protection: masks, respirators (dust, Chemicals)

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Task	Questions Description	Marks
3	Explain the inter-relationship between workplace safety and a productive work environment	7

- Safety boots productivity : When employees Feels Safe , thay're more Focused and efficient

- less accidents = less downtime : Fewer injuries means less time off and Fewer disruptions

- Morale and motivation : A Safe workplace improves job satisfication and reduces Stress

- less downtime and Delays
Unsafe environmets can cause equipment damage, injuries , or investigations that Stop work

- improved work Quality
Employees who work in a safe environment can concentrate better on their tasks

- Compliance with laws and regulation
Following workplace safety laws and regulations prevents legal penalties and fines

- Better teamwork and Communication
Safety programs often require communication and cooperation between workers and supervisors

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Task	Questions Description	Marks
5	What are the risk of fire and explosion	8

Fire hazards become fire risks if the hazard is likely to have significant impact on people and property

- ~~injury or death~~ A fire requires three elements to coexist simultaneously. If you remove one the Fire goes out
- Fuel : Combustible materials (wood, paper gases, Chemicals)
- oxygen : Usually from the surrounding air
- Heat : An ignition source like a spark, Friction or open Flame

Explosion - ~~An explo~~

- Dispersion : The Fuel (like dust or gas) is spread out in the air
- Confinement : The Mixture is trapped in an enclosed space (like a room or a pipe) allowing pressure to build up.

Task	Questions Description	Marks
6	Functions of approve inspection authorities for electrical installations	12

1. Verification and Compliance Audits
- Design Review : Assessing electrical plans before construction to ensure the proposed layout is safe
- Physical inspection : Checking wiring, earthing (grounding) and

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Common types of Earthing

- Plate earthing - using a metal plate buried in soil
- Rod earthing - using a metal rod driven into the ground
- Pipe earthing - using a metal pipe in the ground

How Earthing Works

* A metal rod or plate is buried deep in the ground
* A thick earthing wire connected this rob to electrical equipment or to distribution board.

Task	Questions Description	Marks
8	Why lockout/tag out important?	4
	What is the purpose of a lockout/tag?	3
	What does de-enegization mean?	3
	What should lockout procedures and work instruction contain?	10

Lock out / tagout (LOTO)

- Worker Safety
- injury Prevention
- Assest Protection
- Legal Compliance

Prop Purpose of a lockout / tag

- Lockout (Physical Barrier) The primary purpose is to prevent the equipment from being energized or started
- Tagout (Communication) - The tag acts as a Warning System
- Energy Isolation

- De-energization is the process of removing all energy sources from a system. It specifically involves
- Disconnecting the sources
- Releasing Stored energy
- Zero - energy state

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Task	Questions Description	Marks
9	Verify isolation explain the several ways to be followed	8

* Identify the correct circuit
- Identify the correct circuit or equipment that needs to be isolated
- Check the distribution board, switch or breaker that supplies power to that equipment
* Switch off the power supply
- Turn off the circuit breaker, isolator, or main switch that supplies electricity to the circuit
* lockout and tagout
- Apply a lock on the switch or breaker so no one can turn it on accidentally
- place a warning tag saying "Do not switch on - maintenance in progress"
* Testing the voltage Tester
Before testing the circuit, test your voltage tester or multimeter on a known live source to ensure the tester works properly
* Test the Circuit
Use the tester to check between
- Live and neutral
- Live and Earth
- Neutral and Earth
* Re-test the Tester
after testing the circuit, check the tester again on live source to confirm it is still working

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SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	Legisiation and regulations for workplace safety within electricity			✓
2	Safety precautions and safe practices for working within industry			✓
3	Personal protective equipment			✓
4	Fundamentals of isolating and locking out equipment and circuits			✓

S. _____ _____

Learner Signature **Date**

_____ _____

Facilitators Signature **Date**

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DECLARATION BY LEARNER			
I, ... Sela Phiri .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... Ruth Nyakala ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

Notification of Assessment

Assessment will take place as follows:

	Theory	**Practical**
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONE

Assessment Planning

	Theory	**Practical**
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

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